DATA PRODUCT CASE STUDY

PRIME: one governed data layer for a multi-tenant product

Then the same approach, live on City of Calgary transit data.

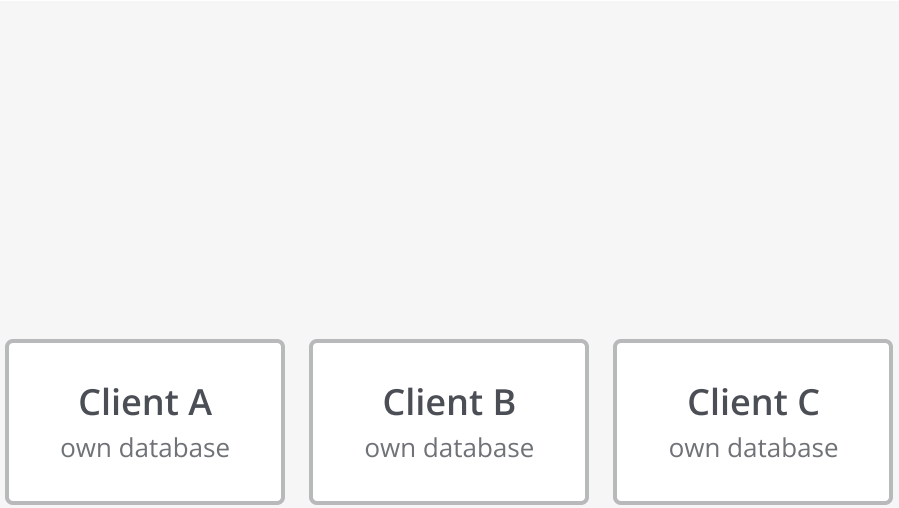
Cody Chandler
Data and Business Analyst interview · The City of Calgary

I'll answer them in order

1. What business problem was I solving?
2. What is the data product?
3. How did I approach it?
4. What challenges came up?
5. What am I most proud of?
6. What would I improve with more time?

With a 5-minute live demo on City of Calgary data after question 3.

What business problem was I solving?



The problem

A multi-tenant product kept each client's data in its own database, so internal reporting was stale (up to 7 days old), fragmented and unowned.

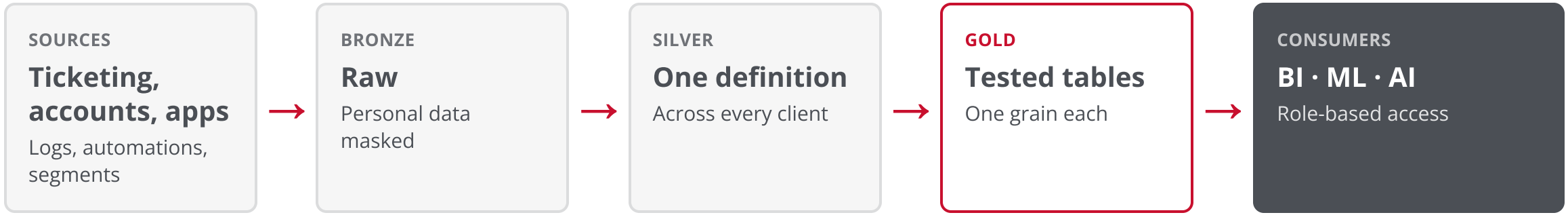
a Why it mattered

Teams and clients worked from different numbers, and every cross-client question needed engineering.

b Who it affected

Customer Success · Product · Clients · Analysts · AI assistants

What is the data product?



a What it does

Consolidates every client into one governed, tested layer, refreshed daily, that every team and tool reads from.

b How people use it

- BI dashboards
- Client data sharing
- CS platform
- Plain-English questions

C What value it delivered

~~7 days~~

< 12 hrs

Data freshness

100+

Tested models, replacing a database per client

30 min

To flag a missed refresh

Self-serve

Routine Customer Success questions

When the dashboard, the client export and the AI give the same number, people stop checking and start deciding.

How did I approach it?

Problem → Needs → Strategy → Plan → Build → **Review and iterate**

a **Sourcing and preparing the data**

Six source types (ticketing, accounts, app logs, automations, segments, reference files), mapped client by client into one model.

b **Methods and tools**

Snowflake · dbt · Airflow · AWS · Medallion layers · Data contract · Semantic layer · LLMs

The same approach, live on City of Calgary data

1. Dashboard: riders back to 95% of 2019
2. The catch: a 43% jump that wasn't real
3. Predict: a live XGBoost model
4. Ask the data: plain English, with its working

Airflow · dbt · DuckDB · 27 tests · XGBoost · AI agent · codychandler.ca/data-demos/calgary-transit

What challenges came up, and how did I handle them?

Type	Challenge	How I solved it
Technical	Every client system had its own shape.	One dimensional model all clients map into.
Analytical	The same metric meant different things to different teams.	One definition in silver, one documented grain per gold table.
Interested parties	Nobody owned freshness or accuracy.	A data contract with named owners, targets and tiers.

What am I most proud of?

Governance stopped being overhead and became the reason people could self-serve.

One tested layer fed dashboards, client data and an AI assistant, with no separate pipeline for each. The answer you just saw is that, working.

What would I improve with more time?

PRIME for clients

Take the internal CS bot to clients as Stellar Insights.

On-time performance

Add the City's real-time feeds to measure reliability by route.

Owner sign-off

Agree targets and definitions with the people who run the service.

The same work at the City: find where numbers disagree, agree one definition with the owners, and make it self-serve.

Thank you. Questions?

Live demo	**codychandler.ca/data-demos/calgary-transit**
Code	**github.com/frontier-platforms/cody-portfolio**